

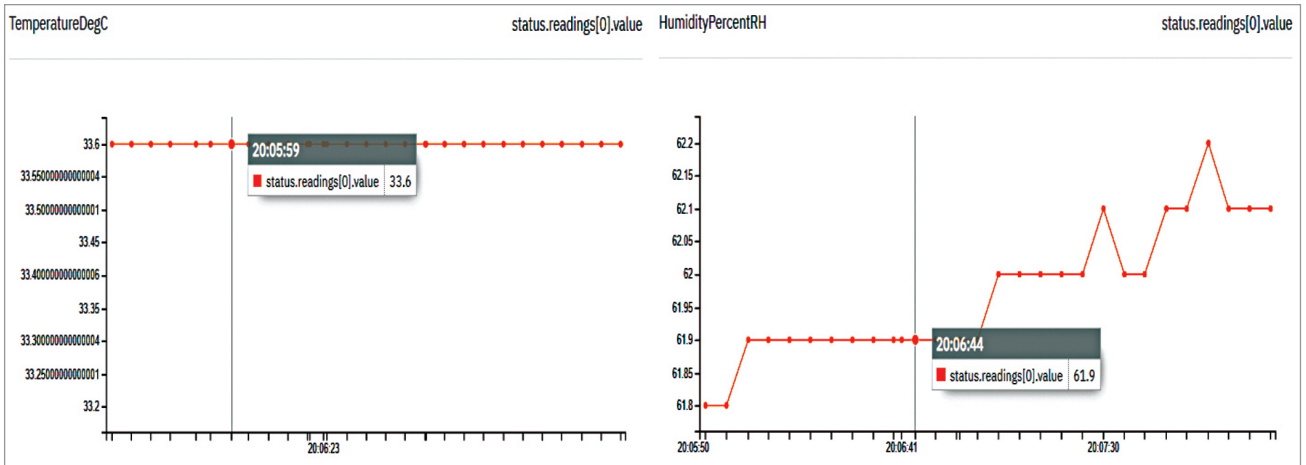


Fig. 9: Sensor data snapshots from IBM Cloud portal

bound to southbound (application service to LED actuation):

- Application service sends the sensor data to rules engine based on the configuration. Kuiper is the rules engine.

- The required rules should be added to the rules engine prior to receiving the sensor data from application service, so that it can actuate using the core commands of the GPIO device service mentioned in the rules.

- LEDs that must go on/off are already configured in GPIO device service. For this case study, pin GPIO-12 is configured with red LED and pin GPIO-14 is configured with blue LED. Core commands of GPIO-12 and GPIO-14 are added with corresponding rules in Kuiper.

- Threshold values for case study are, blue LED must glow if the temperature goes below 28 degrees and red LED must glow if the temperature goes above 30 degrees. Between 28 degrees and 30 degrees both the LEDs should be off.

- Sensor data received from application service is analysed by Kuiper and, as per the configured rules, core commands of GPIO device service are actuated when threshold values are reached.

- Based on the commands triggered by Kuiper, GPIO device service

actuates the corresponding LEDs on or off.

Analysing the data flow

Following section analyses the data flow at various stages, when sent from southbound to northbound and vice versa.

Modbus device service. It retrieves the temperature and humidity values from the sensor using Modbus messages every 5 seconds. Modbus logs from the case study are:

```
-----
2021/03/02 14:35:21 modbus:
sending 01 04 00 01 00 01 60 0a
2021/03/02 14:35:21 modbus:
received 01 04 02 01 50 b8 9c
-----
```

The messages sent to and received from the sensor are decoded as shown below:

Sent	Decoding
01	Slave ID
04	Function, read_input_registers
00 01	Register address
00 01	No. of registers to read
60 0a	CRC

Received	Decoding
01	Slave ID
04	Function, read_input_registers
02	Bytes count
01 50	Data, temperature
b8 9c	CRC

Temperature value received is 0x0150, which is 336 in decimal. SHT20 sends the values in multiple of 10, so temperature is 33.6 degrees.

Core and application services

Sensor data values captured at core service using REST APIs of core command are shown in Fig. 8.

Application service will send the data to IBM Cloud, which is configured in the configuration.toml file. Application service logs when data is sent to the cloud are:

```
-----
level = DEBUG ts = 2021-03-21T14:35:48.355521757Z
app = AppService- source = runtime.
go:59 msg = "Processing message: 5 Transforms"
level = DEBUG ts = 2021-03-21T14:35:48.36118818Z
app = AppService- source = filter.
go:44 msg = "Filtering by DeviceID"
level = DEBUG ts = 2021-03-21T14:35:48.364375438Z app = AppService- source = filter.go:86
msg = "Filtering by ValueDescriptor"
level = DEBUG ts = 2021-03-21T14:35:48.367788325Z app = AppService- source = conversion.
go:62 msg = "Transforming to JSON"
level = DEBUG ts = 2021-03-21T14:35:48.371520359Z app = AppService- source = mqtt.go:138
-----
```